Homework 2

Problem 1:

Use the Cauchy-Riemann conditions to find out whether the functions in (1)~(4) are analytic.

(1)

$$\frac{2z-i}{iz+2}$$

(2)

$$z^2\bar{z}$$

(3)

$$\sqrt{z}$$

(4)

$$e^{iz} \quad (\textit{Careful:} \quad \cos z \text{ and } \sin z \text{ are } \textit{not } u \text{ and } v.)$$

Problem 2:

Show that the following functions are harmonic, that is, that they satisfy Laplace's equation, and find for each a function $f(z)$ of which the given function is the real part. Show that the function $v(x, y)$ (which you find) also satisfies Laplace's equation.

(1)

$$3x^2y - y^3$$

(2)

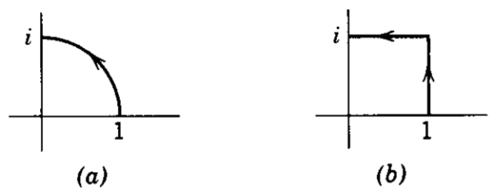
$$e^x \cos y$$

(3)

$$\ln(x^2 + y^2)$$

Problem 3:

$\int_1^i z\,dz$ along the indicated paths:



Problem 4:

If $f(z)$ is analytic on and inside the circle $|z| = 1$, show that $\int_0^{2\pi} e^{i\theta} f(e^{i\theta})\, d\theta = 0$.

Problem 5:

Use Cauchy's theorem or integral formula to evaluate the integrals

$\oint_C \frac{\sin z\, dz}{2z - \pi}$ where C is the circle $\begin{matrix}\text{(a) } |z| = 1, \\ \text{(b) } |z| = 2.\end{matrix}$

Problem 6:

Use Cauchy's theorem or integral formula to evaluate the integrals

$\oint_C \frac{\sin 2z\, dz}{6z - \pi}$ where C is the circle $|z| = 3$.

Problem 7:

Use Cauchy's theorem or integral formula to evaluate the integrals

$\oint \frac{e^{3z}\, dz}{z - \ln 2}$ if C is the square with vertices $\pm 1 \pm i$.

Problem 8:

Use Cauchy's theorem or integral formula to evaluate the integrals in (1)-(4).

(1)

$$\oint_{|z|=2} \frac{z^2 - 1}{z^2 + 1} \mathrm{d}z;$$

(2)

$$\oint_{|z|=2} \frac{\sin\left(\mathrm{e}^z\right)}{z} \mathrm{d}z;$$

(3)

$$\oint_{|z|=2} \frac{\sin z}{z^4} \mathrm{d}z;$$

(4)

$$\oint_{|z|=2} \frac{\mathrm{d}z}{z^2(z^2 + 16)}.$$